Thin-fiber monodromy, global reduction, and Einstein realization

24 September 2026. This note removes the reference-fiber thickness hypothesis from the preceding cycle theorem, at the cost of an explicit alternative: finite monodromy or a common invariant essential multicurve. For irreducible monodromy the alternative gives exact negative Kähler–Einstein existence on the same smooth total space. For reducible monodromy it gives an actual decomposition of the surface bundle, not an assertion that its pieces already have compatible Einstein metrics.

The proof uses classical hyperbolic collars, quasiconformal length distortion, and compactness of thick moduli. It does not invoke an unverified Teichmüller Margulis theorem or claim novelty for these classical facts. The project-level extension is their composition with the actual cycle budget and pointwise Ricci input. The previous cycle theorem and Ricci theorem are inputs. Their finite-monodromy realization is retained, not rediscovered.

1. Statements with original geometric inputs

Let

$$F_f \longrightarrow M \xrightarrow{\pi} B_b, \qquad f, b \geqslant 2, \tag{1}$$

be a smooth oriented bundle of closed oriented surfaces. Its monodromy is $\rho : \pi_1(B, b_0) \to \mathrm{Mod}(F_f)$. A multicurve means a finite nonempty set of distinct isotopy classes of disjoint essential simple closed curves. Call the image of ρ **irreducible** when it preserves no such multicurve setwise. This is a property of the entire group; it is not a claim that each element is pseudo-Anosov or that the image is infinite.

For any smooth metric g use its orthogonal Ehresmann connection. For a loop γ based at b_0, let Φ_t denote actual horizontal transport and set

$$h_t = \Phi_t^*(g|_{F_{\gamma(t)}}), \qquad A_t = h_t^{-1}\dot{h}_t, \qquad A_t^0 = A_t - \tfrac{1}{2}\,\mathrm{tr}(A_t)I.$$

For a finite based generating system Γ define

$$\mathscr{J}_g(\gamma) = \frac{1}{2}\int_0^1 \sup_{F_{b_0}} \|A_t^0\|_{\mathrm{op}}\, dt, \qquad \mathscr{J}_g(\Gamma) = \max_{\gamma\in\Gamma} \mathscr{J}_g(\gamma). \tag{2}$$

All loops in Γ have the same base point. No representative chosen merely to make the mapping-class displacement small replaces the transport in (2).

Theorem A (global thin-fiber alternative). There is a constant $\mu_f > 0$, depending only on fiber genus, such that

$$\mathscr{J}_g(\Gamma) < \mu_f \tag{3}$$

has the following consequence. Either the monodromy image is finite, or it preserves a multicurve C with at most $3f-3$ components. In the latter case C can be chosen so that every component has length less than 1 in the curvature -1 uniformization of the reference fiber.

If the monodromy is irreducible, it therefore has finite image and the same smooth M carries a metric $\widehat{g}$ with

$$\mathrm{Ric}_{\widehat{g}} = -\widehat{g}, \qquad (\widetilde{M}, \tilde{\widehat{g}}) \cong \mathbb{H}^2_{-1} \times \mathbb{H}^2_{-1}. \tag{4}$$

The output is Kähler in both orientations. There is no reference systole assumption, no prescribed Einstein background, and no claim that the input metric is already Einstein or close to the output. The constant is independent of base genus and generator count.

The alternatives need not be disjoint: a finite group can preserve a multicurve. The assertion relevant to an infinite image is that one **common** multicurve is preserved by every element, not merely that each generator has its own reduction system.

Theorem B (pointwise Ricci realization without a thick fiber). Suppose $\pi : (M, g) \to (B, k)$ is a Riemannian submersion with minimal fibers. Intrinsic fiber curvature is unrestricted and the horizontal distribution need not be integrable. Define

$$\delta = \| \operatorname{Ric}_g^{\sharp} + I \|_{\infty,\mathrm{op}}, \quad \eta = \| (K_k + 1)_+ \|_{\infty}, \quad D_B = \operatorname{diam}(B, k). \tag{5}$$

If

$$2D_B \sqrt{\delta + \eta} < \mu_f, \tag{6}$$

then the finite-or-reducible conclusion of Theorem A holds. In particular, if ρ is irreducible, (4) holds on the original smooth total space. For a hyperbolic base $K_k = -1$, this is the sufficient condition

$$\delta < \frac{\mu_f^2}{4D_B^2}. \tag{7}$$

Consequently every such submersion with **infinite irreducible** monodromy satisfies the input obstruction

$$(\delta + \eta) D_B^2 \geqslant \frac{\mu_f^2}{4}. \tag{8}$$

This concerns the specified submersion metrics, not arbitrary metrics on M. In the irreducible small-input case it nevertheless proves the exact Einstein existence requested by Yau26 for that input class. No total-space diameter, volume, injectivity-radius or auxiliary monopole assumption is used.

2. A simultaneous short-curve lemma

On Teichmüller space use $d_T(x, y) = \frac{1}{2} \inf \log K$. For a curve class c, write $\ell_x(c)$ for its hyperbolic geodesic length.

Two classical facts enter the proof. First, the collar theorem implies that all simple geodesics of length at most 1 are pairwise disjoint. Indeed the collar inequality for intersecting simple geodesics gives $\sinh(\ell(c)/2) \sinh(\ell(d)/2) > 1$, whereas $\sinh^2(1/2) < 1$. Thus there are at most

$$N_f = 3f - 3 \tag{9}$$

such classes. Primary sources are Keen, *Collars on Riemann surfaces*, and Buser, *The collar theorem and examples.*

Second, Wolpert's quasiconformal length inequality gives

$$e^{-2d_T(x,y)} \ell_x(c) \leqslant \ell_y(c) \leqslant e^{2d_T(x,y)} \ell_x(c). \tag{10}$$

See *The length spectra as moduli for compact Riemann surfaces.* The needed normalization can also be checked directly. The cyclic cover corresponding to c, with core length ℓ, is a conformal annulus of modulus π/ℓ. A K-quasiconformal map lifts to the corresponding cyclic covers and changes

annular modulus by a factor between K^{-1} and K. Take the infimum over marked maps to obtain (10). This argument applies to every essential simple class, not just a chosen pants decomposition.

Put

$$a_f = \frac{\log 2}{N_f + 1} = \frac{\log 2}{3f - 2}, \qquad \nu_f = \frac{a_f}{4}. \tag{11}$$

Lemma 1 (a common reduction, with no generator-count bound). Suppose $\mathrm{sys}(x) < 1/2$, and let S be any set of mapping classes satisfying

$$d_T(x, sx) < \nu_f \quad (s \in S). \tag{12}$$

Then $\langle S \rangle$ preserves a nonempty multicurve all of whose x-lengths are less than 1.

Proof. In the interval $[-\log 2, 0]$, list the logarithms of the lengths in $[1/2, 1]$. There are at most N_f entries, including repetitions if distinct curves have equal lengths. Insert the two endpoints. Among the resulting at most $N_f + 1$ gaps there is one (u, v) with

$$v - u \geqslant a_f. \tag{13}$$

There is no curve length with logarithm in this open interval. Let $t = \exp((u + v)/2)$ and define

$$C = \{c : \ell_x(c) < t\}. \tag{14}$$

It is nonempty because the systole is less than $1/2$, and it is a multicurve by the collar theorem. Every member of C has length at most e^u. The action convention may invert s in (10), but $d_T(x, s^{-1}x) = d_T(x, sx)$, so

$$\ell_x(sc) \leqslant e^{2d_T(x,sx)} \ell_x(c) < e^{a_f/2} e^u \leqslant e^{(u+v)/2} = t. \tag{15}$$

Hence $sC \subset C$. Since s is a bijection of curve classes and C is finite, $sC = C$. The same C works for all generators, and is therefore preserved by their entire group. Repeated length values and curves lying at the endpoints of the gap cause no problem: the strict inequality in (12) supplies the strict inequality in (15). □

This is a length-gap argument, not an assertion that small moves of the generators make every group element a small move. Long words may accumulate arbitrarily large displacement.

Lemma 2 (thick finite-isotropy neighbourhood). Some $\varepsilon_{f,1/2} > 0$ has the property that, if $\mathrm{sys}(x) \geqslant 1/2$ and $d_T(x, sx) < \varepsilon_{f,1/2}$ for every $s \in S$, then $\langle S \rangle$ is finite.

Proof. Mumford compactness gives a compact set in Teichmüller space meeting every orbit with this systole bound. If the assertion failed, conjugate to obtain $x_j \to x$ in that compact set and sets S_j generating infinite groups, with all displacements less than $1/j$. Uniformly in $s \in S_j$,

$$d_T(x, sx) \leqslant 2d_T(x, x_j) + 1/j \longrightarrow 0. \tag{16}$$

Properness of the mapping-class action and properness of the finite-dimensional Teichmüller metric imply that the set of mapping classes moving x by at most 1 is finite. Its positive displacements have a positive minimum. Thus for large j every member of S_j fixes x. The stabilizer is finite, giving a contradiction. A finite ineffective kernel, such as the genus-two hyperelliptic involution, does not affect this argument. This is the lemma proved in the preceding cycle note, with $\iota = 1/2$. The classical compactness source is Mumford, *A remark on Mahler's compactness theorem.* □

Fix once and for all

$$\mu_f = \min\{\varepsilon_{f,1/2}, \nu_f\} > 0. \tag{17}$$

Lemmas 1–2 give the following purely geometric alternative for every $x \in \mathscr{T}_f$: the subgroup generated by any collection of μ_f-small moves is finite or has a common short multicurve. The second conclusion is not replaced by "each generator is reducible."

3. From the actual connection to the group alternative

For completeness the differential-to-discrete step has no compactness hypothesis. For two fixed nonzero tangent vectors on the reference fiber,

$$\left|\frac{d}{dt}\log\frac{h_t(v,v)}{h_t(w,w)}\right| \leqslant 2\|A_t^0\|_{\mathrm{op}}. \tag{18}$$

Integration bounds the logarithm of the quasiconformal dilatation of the identity between h_0 and h_1 by $\int \sup \|A_t^0\|_{\mathrm{op}}\, dt$. The endpoint metric is the pullback by the actual return map, which represents $\rho(\gamma)$. Therefore

$$d_T(x_0, \rho(\gamma)x_0) \leqslant \mathscr{J}_g(\gamma). \tag{19}$$

Noncommuting shape operators and a curved connection are allowed. The mapping classes of return maps are invariant under based homotopy and compose as monodromy, even though the actual return diffeomorphisms need not form a representation.

Apply (17) to (19) for the generating system. This proves the alternative in Theorem A. If ρ is irreducible, the short multicurve is impossible, so the image G is finite. Kerckhoff's Nielsen realization theorem realizes G by orientation-preserving isometries of one hyperbolic fiber. On the connected kernel cover $\widehat{B} \to B$, use a lifted hyperbolic base metric. The diagonal action on $\widehat{B} \times F_f$ is free, regardless of fixed points of the fiber action, since its base action is free. The product metric descends with Ricci tensor equal to minus the metric.

This construction gives the original smooth bundle, not just a manifold with an isomorphic group. For $f \geqslant 2$, Earle–Eells contractibility of $\mathrm{Diff}_0^+(F_f)$ gives $B\,\mathrm{Diff}^+(F_f) \simeq B\,\mathrm{Mod}(F_f)$. The two bundles with the same monodromy are therefore smoothly bundle-isomorphic. Pull back the product-quotient metric. The product complex structures with either sign on the fiber descend, giving (4) in both orientations. This concludes Theorem A. The primary contractibility source is Earle–Eells, *The diffeomorphism group of a compact Riemann surface.*

4. Ricci input and the scale of irreducible monodromy

For a horizontal vector X, write

$$g(S_X U, V) = \langle (\nabla_U V)^{\mathscr{H}}, X\rangle, \qquad A_X Y = \tfrac{1}{2}[X,Y]^{\mathscr{V}}.$$

Minimality gives $\operatorname{tr} S_X = 0$. The classical horizontal Ricci identity, already used in retained Round 20, is

$$\mathrm{Ric}_g(X,X) = K_k - |S_X|_{\mathrm{HS}}^2 - 2\sum_a |A_X e_a|^2 \quad (|X| = 1). \tag{20}$$

The mean-curvature derivative term vanishes because the fibers are pointwise minimal. The source is O'Neill, *The fundamental equations of a submersion.* Equation (5) consequently implies

$$|S_X|_{\mathrm{op}} \leqslant \sqrt{\delta + \eta}\,|X|. \tag{21}$$

Horizontal transport differentiates the vertical metric by $h_t^{-1}\dot{h}_t = -2\Phi_t^* S_{\dot{\gamma}\mathscr{K}}$. Thus (2) gives

$$\mathscr{J}_g(\gamma) \leqslant \sqrt{\delta+\eta}\ \mathrm{Length}_k(\gamma). \tag{22}$$

For each $a > 0$, a finite based generating system of a closed Riemannian manifold can be chosen with lengths below $2\,\mathrm{diam} + a$. One direct proof is to subdivide any loop into sufficiently short arcs and join the subdivision points to the base point by paths of length at most the diameter. Each resulting loop has length below $2\,\mathrm{diam} + a$. The set of their possible homotopy classes is finite: in the proper universal cover it is represented by deck transformations whose orbit points lie in a fixed compact ball. These classes generate the fundamental group. No bound on their number is needed here.

Under (6), choose $a > 0$ small enough that $(2D_B + a)\sqrt{\delta+\eta} < \mu_f$ and apply Theorem A to (22). If $\delta + \eta = 0$, every cycle budget is zero and the same argument applies directly. This proves Theorem B and its contrapositive (8).

There is also an explicit thickness consequence, without the implicit constant in Mumford compactness. If ρ is irreducible and *some* fiber has conformal hyperbolic systole less than $1/2$, base the short generators there. Lemma 1 and (22) give

$$(\delta+\eta)D_B^2 \geqslant \frac{(\log 2)^2}{64(3f-2)^2}. \tag{23}$$

Hence, below this scale, irreducibility forces every fiber to be conformally thick. This is not a lower bound for four-dimensional ball volumes. It identifies the missing information in the earlier thick-fiber theorem: here a global topological property supplies the fiber thickness needed for finite-monodromy detection.

5. The reducible alternative is a genuine global decomposition

Proposition 3 (fiberwise realization of the invariant multicurve). If ρ preserves a multicurve C of size r, then M contains a smooth submanifold $\mathscr{C}$ whose projection to B is a bundle with fiber the corresponding union of r circles. After a connected finite base cover of degree at most $2^r r!$, its circle components and orientations can be labeled globally. Cutting the pulled-back total space along them gives genuine surface bundles with boundary over that base cover.

Proof. Isotopy classes of curves in the fibers form a discrete local system. The monodromy-invariant set C gives a global section of the associated local system of multicurves. Uniformize the vertical metric in each fiber to curvature -1. These metrics vary smoothly with the base: the conformal curvature equation has an invertible linearization at its negative-curvature solution, and uniqueness makes the local solutions agree.

In each fiber take the unique geodesic representatives of the specified classes. In a local bundle chart the classes are fixed. Their unparametrized geodesic representatives depend smoothly on the hyperbolic metric: the periodic normal Jacobi operator along a closed hyperbolic geodesic has no kernel. The representatives are disjoint, and uniqueness makes the local families agree under changes of chart. Their union is the required embedded smooth $\mathscr{C}$. It is compact and its projection is a submersion, so it is a bundle by local triviality for proper submersions.

The action on oriented components maps $\pi_1(B)$ to the signed permutation group of order $2^r r!$. Pass to the connected cover associated with its kernel. An orientation of a curve, together with the

fiber orientation, orients its normal line. Thus every boundary side obtained by cutting is labeled. Every component of a cut connected surface meets some such boundary side; consequently the components are also fixed. Fiberwise tubular neighborhoods and cutting now produce the stated smooth bundles with boundary. No assertion that the circle bundles are trivial is needed or made. □

This decomposition keeps rather than discards residual monodromy. After the finite cover, let S_v be the cut fiber components and let $\overline{S}_v$ be the punctured surfaces obtained by capping their boundary circles with marked punctures. Restriction defines

$$\rho_v : \pi_1(\overline{B}) \longrightarrow \mathrm{PMod}(\overline{S}_v). \tag{24}$$

For the subgroup fixing curves and their sides, the kernel of the product restriction map is precisely the free abelian group

$$\mathscr{T}_C \cong \mathbb{Z}^r \tag{25}$$

of Dehn twists about C. A representative trivial after capping can be isotoped to the identity off annular neighborhoods; its remaining annular mapping classes are integer twists. Conversely all those twists vanish after capping, and twists in distinct essential nonparallel curves are independent. This is the standard multicurve restriction sequence; see Hubbard–Koch, Proposition 2.5 of *An analytic construction of the Deligne–Mumford compactification of the moduli space of curves*.

Thus the remaining data are the representations (24) **and** the annular twist data (25). If the joint image in (24) is finite, a further finite base cover reduces the image to $\mathscr{T}_C$. Theorems A–B do not prove that additional premise. A common multicurve alone cannot replace it: a mapping class supported on a once-punctured torus with homological action $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ is pseudo-Anosov on that component and extends by the identity across its boundary. It preserves the boundary curve but is not virtually a multitwist. This example explains the retained restriction data; it is not claimed to satisfy the smallness constant in (3).

In particular, neither this reduction nor a local bidisk description solves the Einstein matching problem along $\mathscr{C}$. No invariant multicurve is silently promoted to a global marked product or to finite monodromy.

6. Actual tests of both alternatives

6.1 A nonempty finite irreducible realization class

Finite image is compatible with irreducibility. On the smooth genus-two curve

$$y^2 = x^5 - 1 \tag{26}$$

consider $\phi(x, y) = (\zeta_5 x, y)$. Its action on holomorphic one-forms $dx/y, x\,dx/y$ has eigenvalues ζ_5, ζ_5^2, and its action on $H^1(F; \mathbb{Q})$ has the irreducible characteristic polynomial $\Phi_5(t) = t^4 + t^3 + t^2 + t + 1$.

If a multicurve were preserved, its at most three components would each be fixed by this order-five action. A nonseparating component would give an invariant rational line in H_1, impossible by irreducibility of Φ_5. A separating component divides the genus-two surface into two one-holed tori. The order-five action cannot exchange these two sides, and their homology would give a proper invariant rational two-plane, also impossible. Thus $\langle\phi\rangle$ is irreducible.

Map a surface group onto this cyclic group, take its kernel cover, and form the diagonal product quotient. It satisfies (1) with nontrivial finite irreducible monodromy. The quotient product metric

has $\delta = \eta = \mathscr{J} = 0$. Small, base-dependent vertical metric perturbations with fixed fiber area form, supported over a base chart, give genuinely nonproduct minimal submersion inputs with $\delta \to 0$. For example, exponentiate a smooth trace-free self-adjoint vertical endomorphism with a small base-dependent coefficient, leaving the horizontal and base metrics fixed. The determinant and hence vertical area form stay fixed; the full Ricci tensor varies smoothly. These inputs need not be Einstein, but the conclusion still concerns the same smooth bundle.

6.2 The thin Dehn-twist family realizes the other branch

The retained fixed bundle E_D, with monodromy $\rho(\gamma) = D^{\mu(\gamma)}$ for one nonseparating Dehn twist, provides an actual infinite reducible example. Section 6 of the preceding cycle note gives, on a fixed hyperbolic base and with minimal intrinsically hyperbolic fibers, metrics g_λ such that

$$\delta_-(g_\lambda) = O(\lambda), \qquad \mathscr{J}_{g_\lambda}(\Gamma) = O(\lambda), \qquad \operatorname{sys}(x_\lambda) \leqslant \lambda \to 0. \tag{27}$$

The full Ricci matrix, not merely the horizontal block, was computed there. These metrics also have actual opposite-orientation monopole classes and bilateral near-saturation; they are not formal almost-parallel tensor examples.

Here the common invariant curve is precisely available as the twisting curve. The monodromy remains infinite cyclic however small the budget becomes. With fiber genus f and base genus b, the split fundamental-group extension and the rank-one transvection on fiber homology give

$$b_1(E_D) = 2f + 2b - 1. \tag{28}$$

It is odd, so E_D admits no Kähler structure. Consequently one cannot replace the finite-or-reducible alternative by KE realization for every small-budget bundle. This is not a claim that E_D has no general Einstein metric. The retained example is used as a test, not counted as a new construction.

6.3 A signature-zero infinite irreducible test

Choose a pants decomposition P of F_f and, for each component p, a simple transverse curve q_p supported in the adjacent one-holed torus or four-holed sphere. The collection $P \cup \{q_p\}$ fills: a curve disjoint from P is isotopic to a pants curve, and that class meets its transverse curve. Let G be generated by the twists in this collection.

The group G is infinite and irreducible. Indeed, if it preserved a finite multicurve C, a power of each generating twist would fix every member c of C. The annular intersection formula

$$i(T_a^n c, c) = |n|\, i(a, c)^2 \tag{29}$$

forces $i(a, c) = 0$ for every generating curve a. The filling property rules out such an essential c. Infinitude already follows from one nontrivial twist.

Realize this representation through a free quotient of a base surface group: use one free generator for each twist, send one member of each standard handle pair to that generator and the other to the identity. Actual twist diffeomorphisms give a suspension bundle. The representation extends over a handlebody, so the total four-manifold bounds the associated oriented five-dimensional bundle and has signature zero.

The inherited free-quotient construction in Round 20, Section 2 already constructs $\delta_- \to 0$ metrics of the required minimal hyperbolic-base submersion type on these fixed bundles. No such family is

new here. What (8) newly supplies in the project is the thickness-independent bound

$$\operatorname{diam}(B, k_j) \geqslant \frac{\mu_f}{2\sqrt{\delta_-(g_j)}}. \tag{30}$$

It remains valid even if one tries to degenerate the conformal fibers simultaneously. The earlier signature barrier is zero on these objects; the earlier cycle theorem required a thick reference fiber. Irreducibility detects a global obstruction to small-cycle deformation that both numerical signature and uncontrolled fiber pinching miss. Equation (30) is an obstruction to a bounded-base-scale input, not to Einstein existence on the underlying total manifold.

7. What has changed in the first-line realization mechanism

The previous sufficient condition used a conformally thick fiber to convert small cycle deformation into finite monodromy. The new argument distinguishes two ways of escaping that step. In thin moduli, small simultaneous displacement forces a common short multicurve; in thick moduli it forces finite image. Thus global irreducibility supplies the required thickness rather than assuming it. Under the submersion hypotheses this is a genuine Ricci-input-to-Einstein-output theorem, with a genus-only constant and no fiber systole in the statement.

The reducible case produces a smooth global cutting locus and retains both component and annular monodromy. It is the specific alternative exhibited by the actual E_D near-Einstein family. A further theorem controlling those remaining transition data would be needed for an Einstein realization there. The current conclusion neither presumes that such a theorem holds nor treats the existence problem as settled by local products.

The substantive inputs and their roles are separate: hyperbolic collars and quasiconformal length distortion give Lemma 1; Mumford compactness gives Lemma 2; O'Neill gives (20); Nielsen realization and contractibility of Diff_0 give exact output once finite monodromy has actually been detected. No new gauge solvability or compactness theorem is asserted by this composition.